Classification Analysis

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Agenda

- Supervised Learning
- Classification: Decision Tree
- Regularization
- Classification: Random Forests
- Classification: Adaptive Boosting
- Discussion and Homework

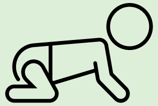
Classification is Supervised Learning aimed at understanding patterns behind the classes

Training

The Data

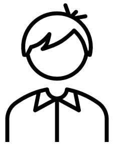
Learning data includes Information about the features of the objects like shape and color.

Input from supervisor: the classes of the objects: Cubes, cylinders…etc.



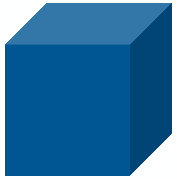
Supervised Training

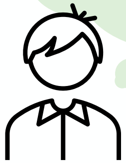
The task is to learn the associations between input features (color and shape) and the provided classes (e.g., cylinders and cubes)



Trained model

Deployment





This is “Cube”, I have seen this pattern before

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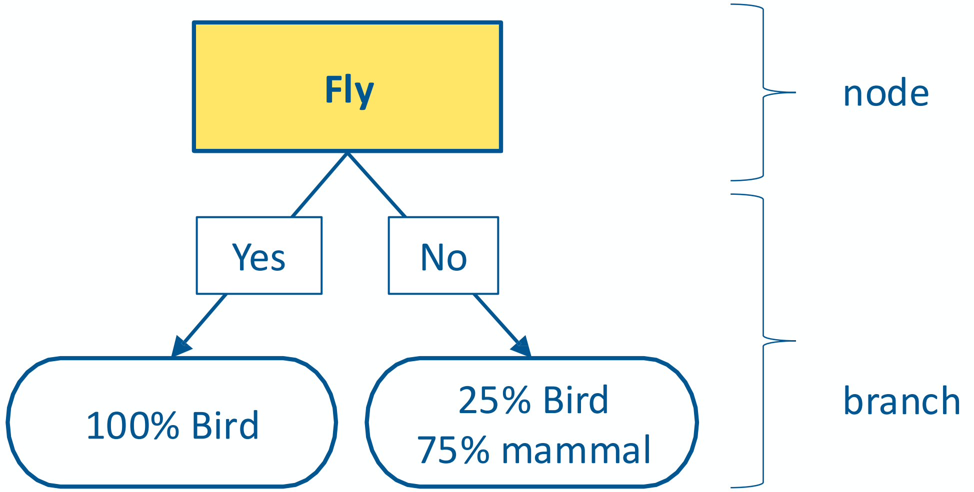
Supervised Learning

Classification with Decision Tree and Random Forests

Decision Tree Learning for classification

- We use labeled data to obtain a suitable decision tree for future predictions
- Each node represents a test on an attribute and each branch is an outcome of that test
- **The goal is to build a decision tree that works well on unseen data, while asking as few questions as possible**

Fly	color	Classification
No	Brown	Mammal
No	White	Mammal
Yes	Brown	Bird
Yes	White	Bird
No	White	Mammal
No	Brown	Bird
Yes	White	Bird



What is a good attribute?

Fly	color	Classification
No	Brown	Mammal
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Yes	Brown	Bird
Yes	White	Bird
No	White	Mammal
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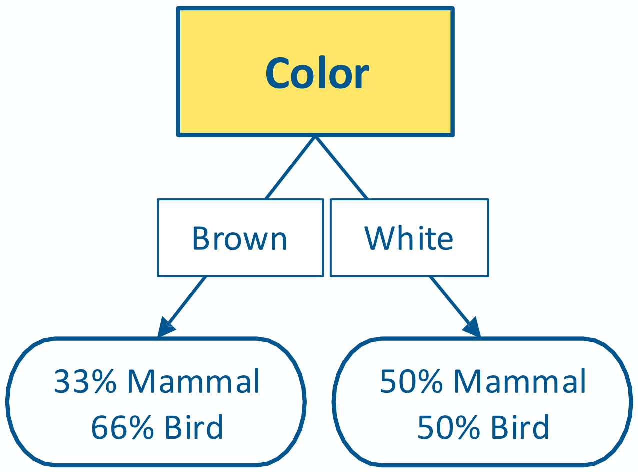
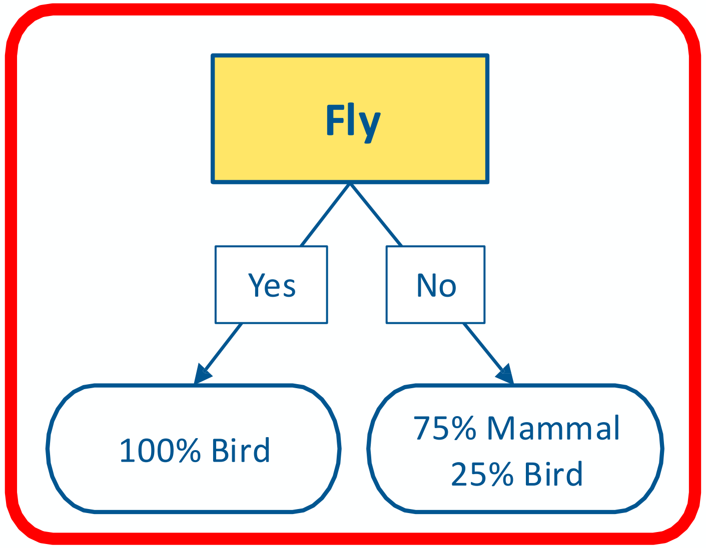


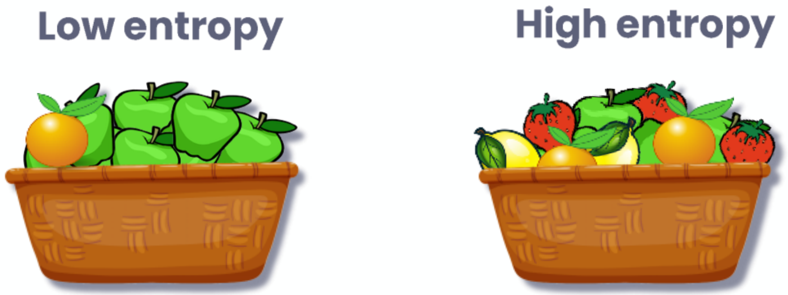


Choose attribute that provides better splitting. But what is better splitting?

- the resulting subsets are more pure (less random and less mixed)
- Knowing the value of this attribute gives us more information about the label

Entropy

- Entropy measures the degree of randomness in data

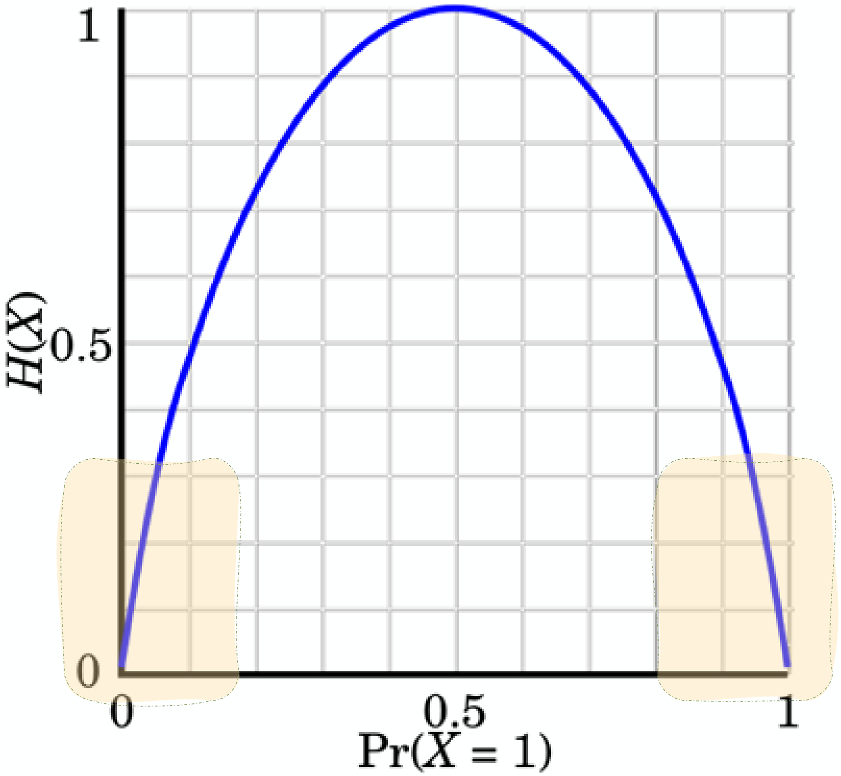


- For a set of samples X with k classes:

$$entropy(X) = -\sum_{i=1}^{k} p_i \log_2(p_i)$$

where p_i is the proportion of elements of class i

- Lower entropy implies greater predictability!



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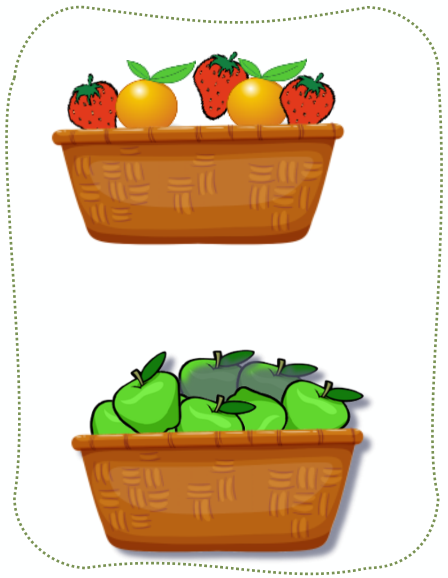
Information Gain

The information gain of an attribute **a** is the expected reduction in entropy due to splitting on values of **a:**

$$gain(X, a) = entropy(X) - \sum_{v \in Values(a)} \frac{|X_v|}{|X|} entropy(X_v)$$

;where X_v is the subset of X for which $a = v$

Split criterion
(color ==Green)



Best attribute = highest information gain

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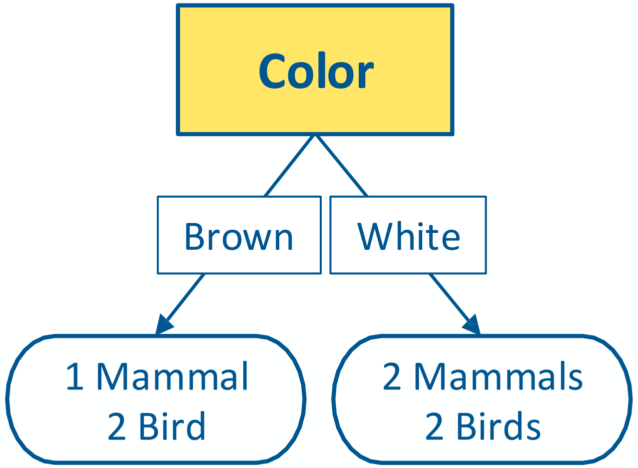
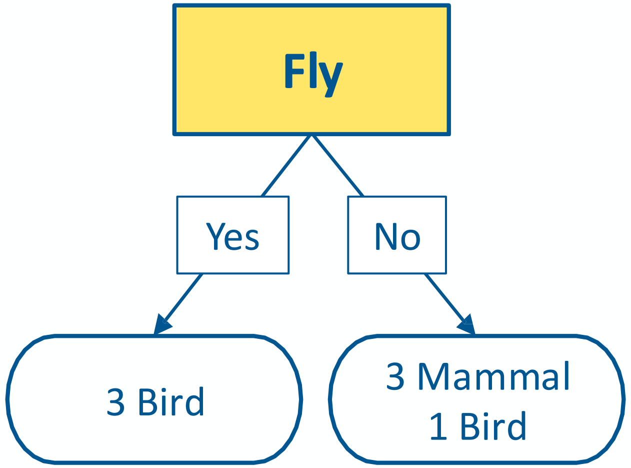




$$entropy\ (X) = -p_{\mathrm{mammal}} \log_2 p_{\mathrm{mammal}} - p_{\mathrm{bird}} \log_2 p_{\mathrm{bird}}$$

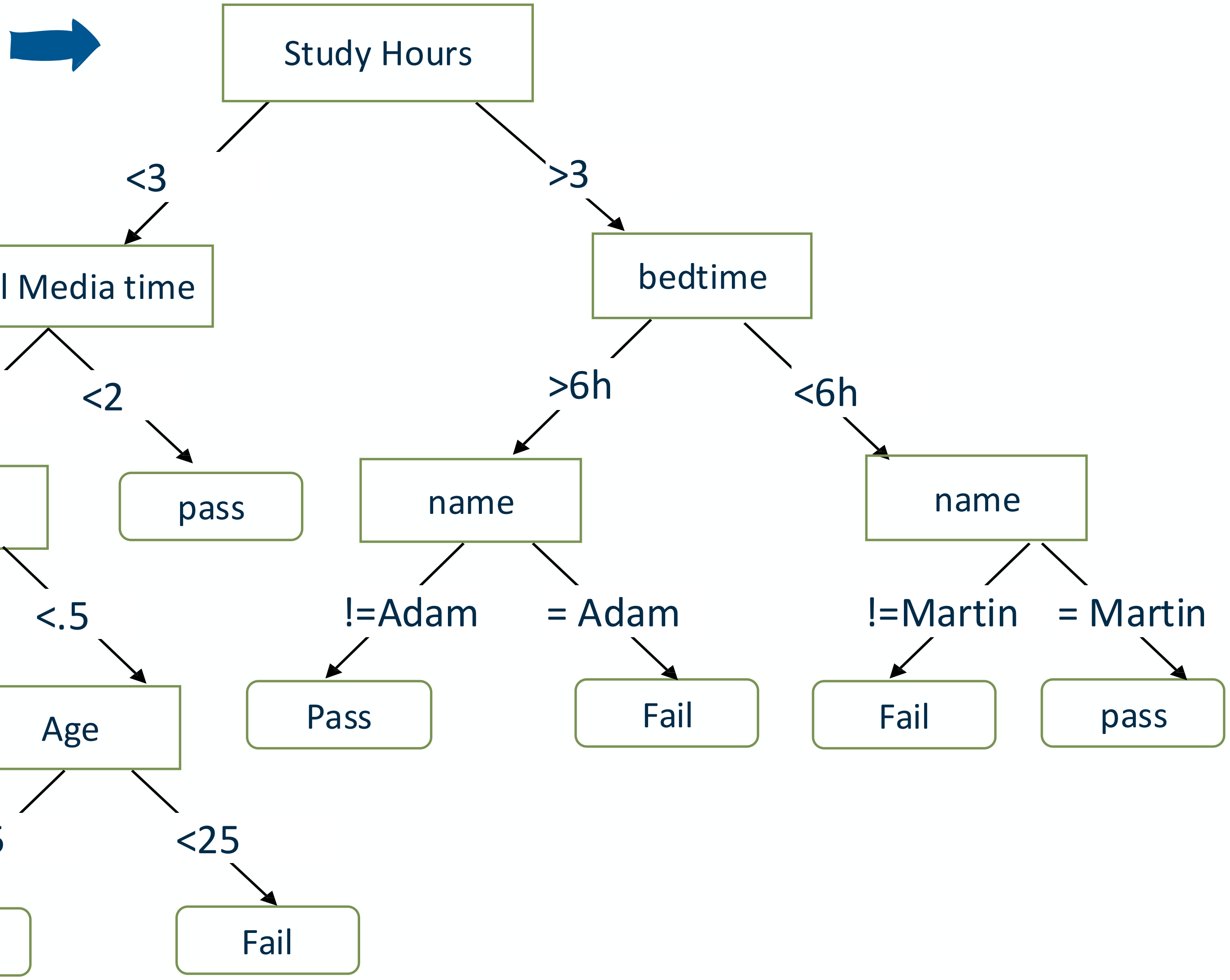
$$gain(X, a) = entropy(X) - \sum_{v \in Values(a)} \frac{|X_v|}{|X|} entropy(X_v)$$

$$\boldsymbol{gain\ (X, color)} = entropy\ (X) - \frac{3}{7} entropy\ (X_{color=brown}) - \frac{4}{7} entropy\ (X_{color=white}) = \mathbf{0.020}$$

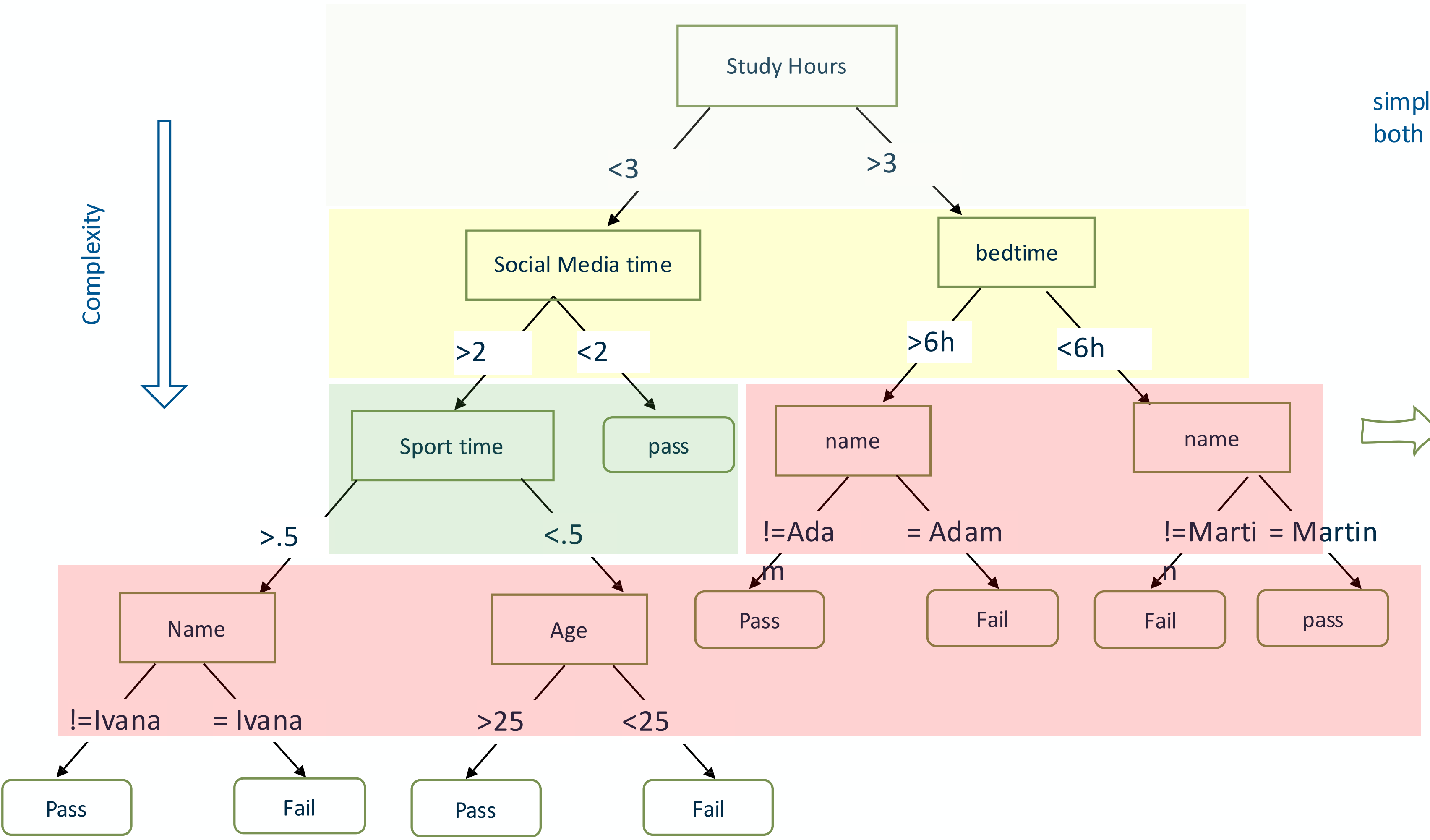
$$\boldsymbol{gain\ (X, fly)} = entropy\ (X) - \frac{4}{7} entropy\ (X_{fly=no}) - \frac{3}{7} entropy\ (X_{fly=yes}) = \boxed{\mathbf{0.521}}$$

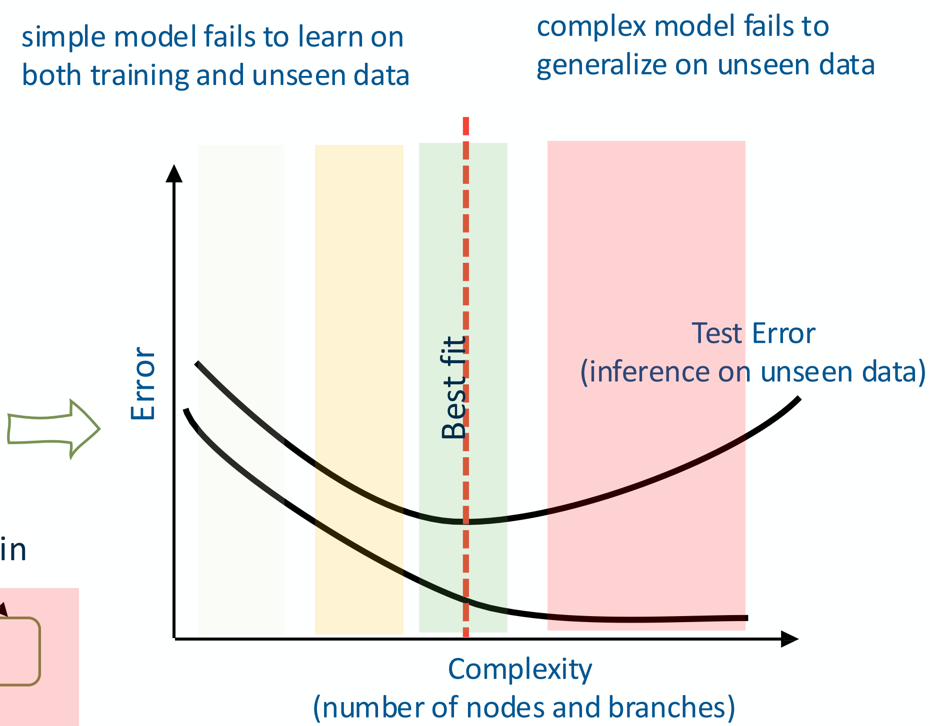
Decision trees & Generalization dilemma

Name	Study Hours	Social Media Time	Sport Time	Bedtime	Age	**Result**
…	…	….	…	…	…	…



Decision trees & Generalization dilemma



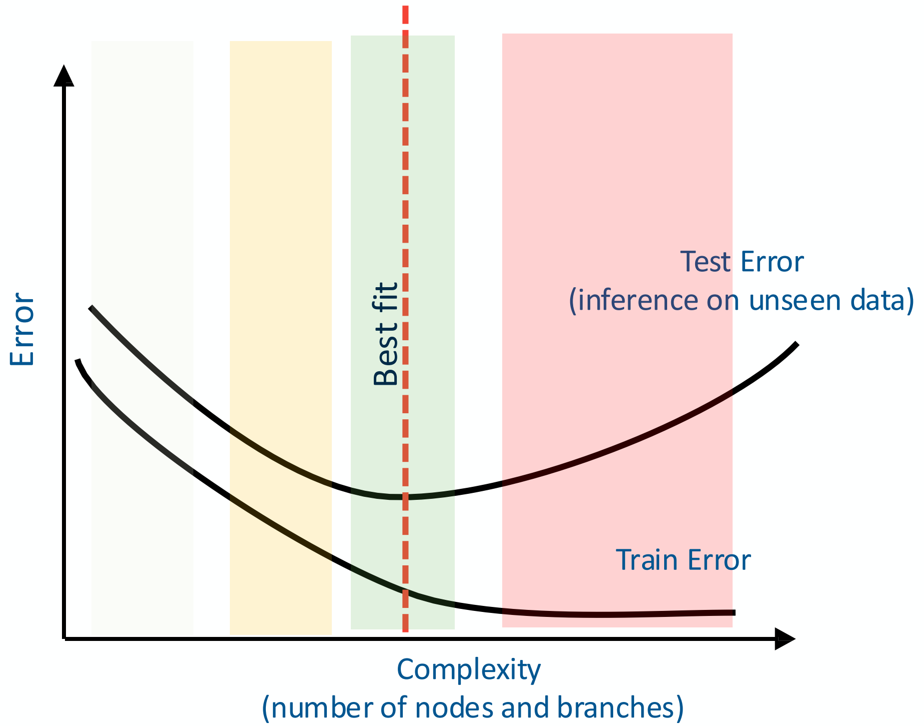


Decision trees & Generalization dilemma

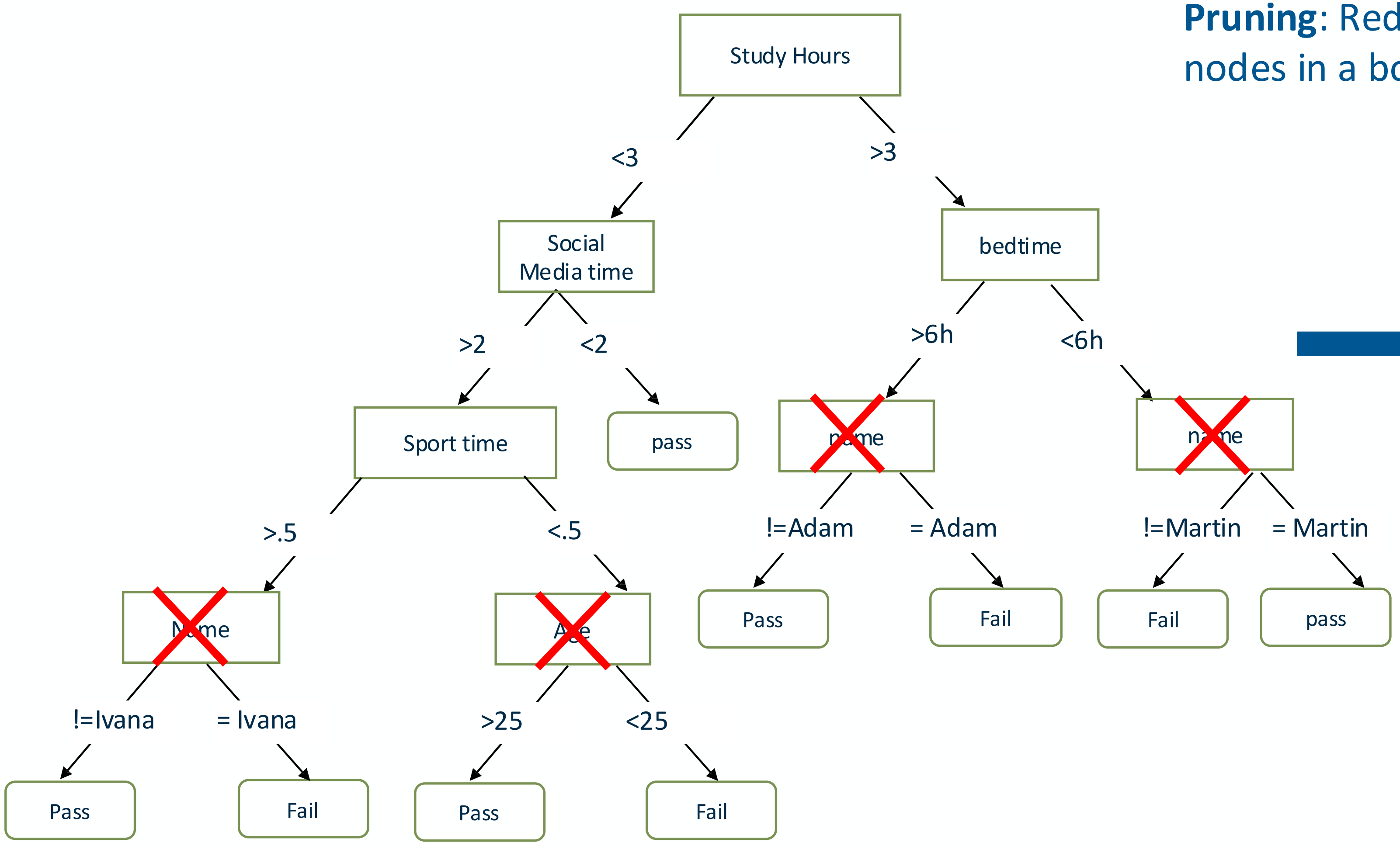
- A good machine learning model generalizes well: Works good for training, and testing data
- A very complex machine learning model may learn noise or very specific information that doesn't necessary exist in unseen data

simple model fails to learn on both training and unseen data

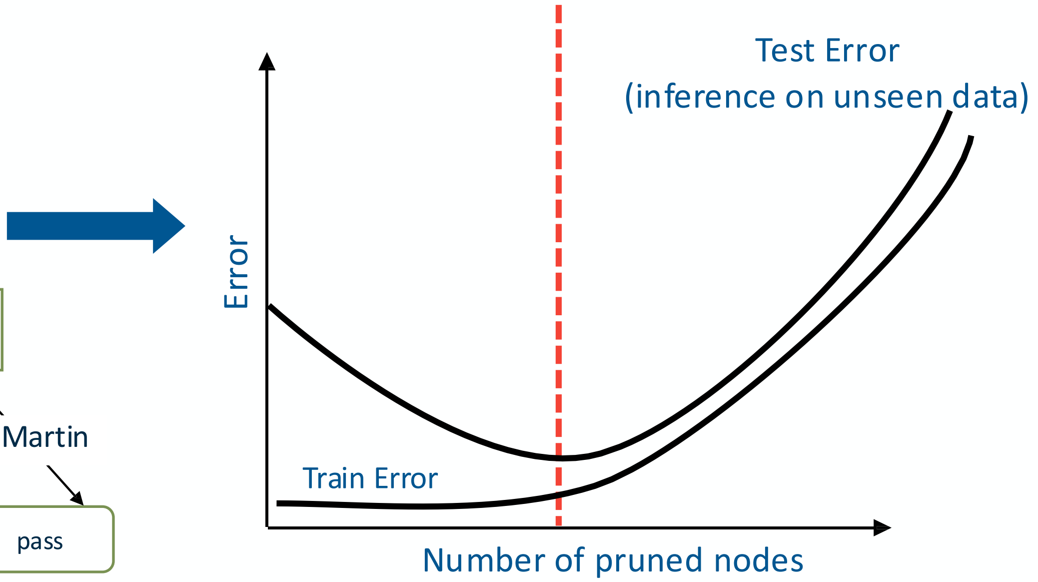
complex model fails to generalize on unseen data



Regularization: Pruning Decision Trees

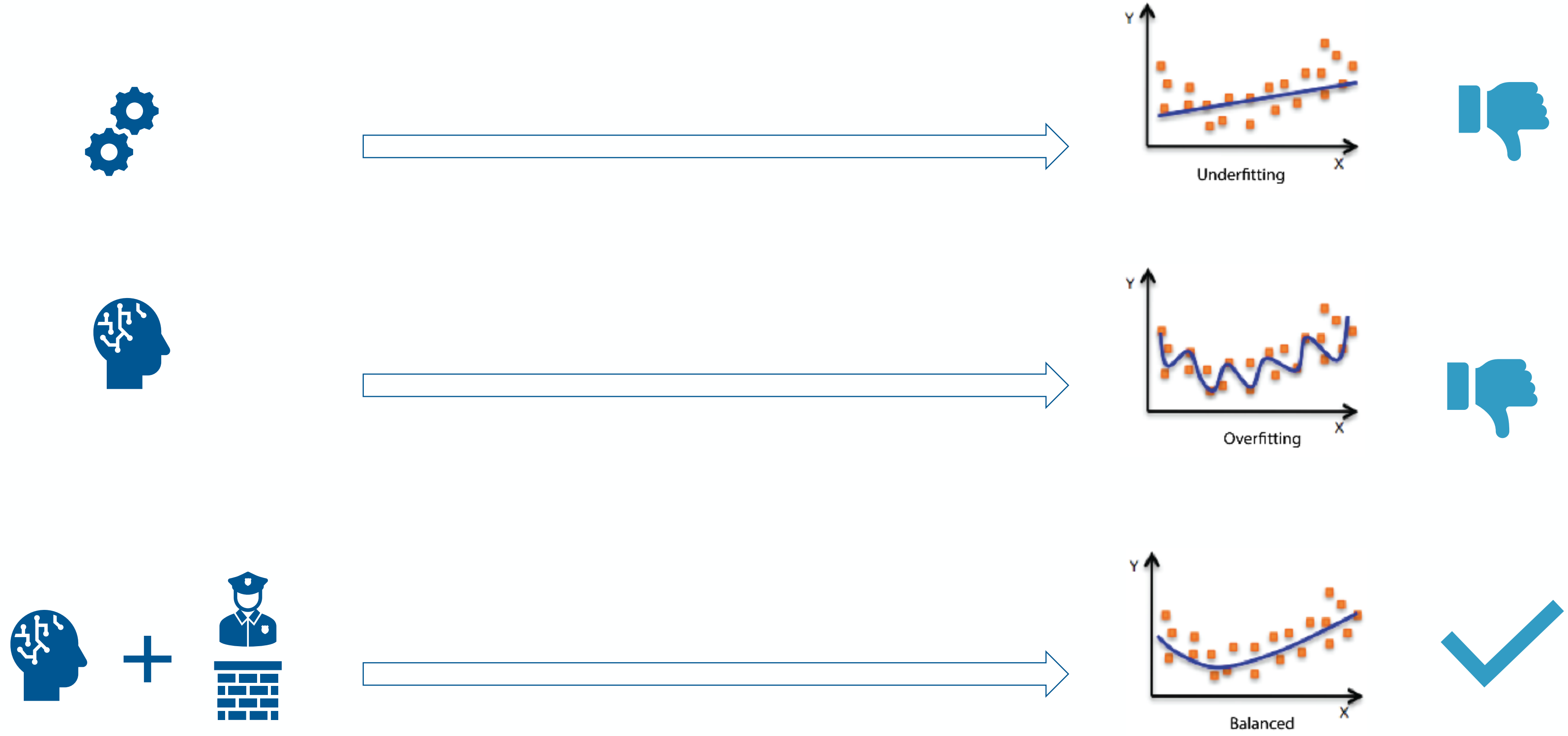


Pruning: Reduce the complexity of the model by pruning nodes in a bottom-up manner, if it decreases validation error.



Regularization in Machine Learning

- **Regularization** is a technique used in machine learning to prevent **overfitting** by adding a **penalty** to the model's complexity.
- It encourages the model to slow down the update of weights (or parameters) to ensure the model is focusing on learning the underlining patterns in the training data making it more generalized and robust on unseen data.
- Otherwise, models may end up learning the training data in a parrot-fashion.



Limitations of Decision Tree

⚠️ **Overfitting Risk**

Tends to create very complex trees that **fit noise** in the data

Especially problematic with small datasets or many features

📉 **Lower Predictive Power (underfitting risk)**

Often **less accurate** than ensemble methods (e.g. Random Forest, XGBoost)

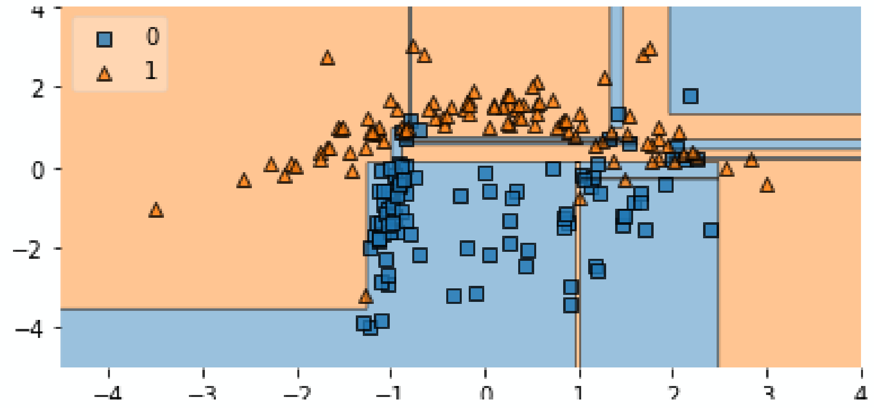
🎲 **Instability**

Small changes in data can lead to a completely different tree structure

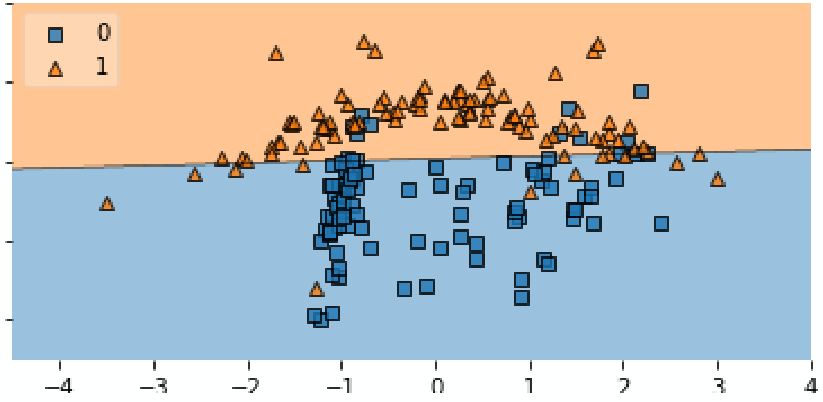
Quiz

- Which one depics the decision boundaries drawn by a decision tree?

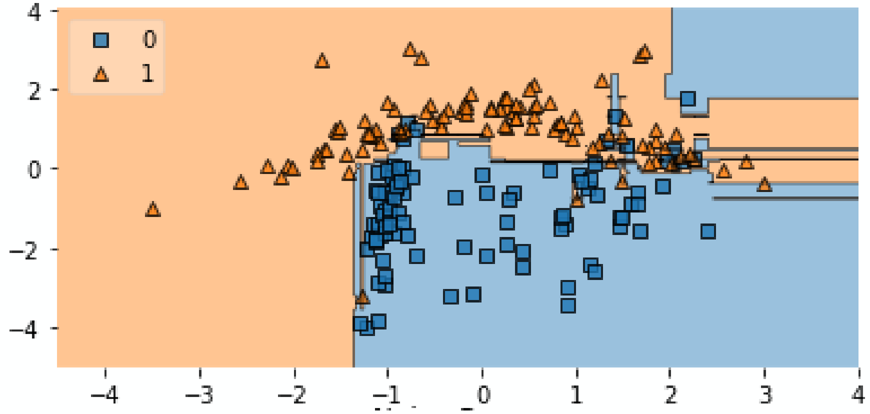
A



C



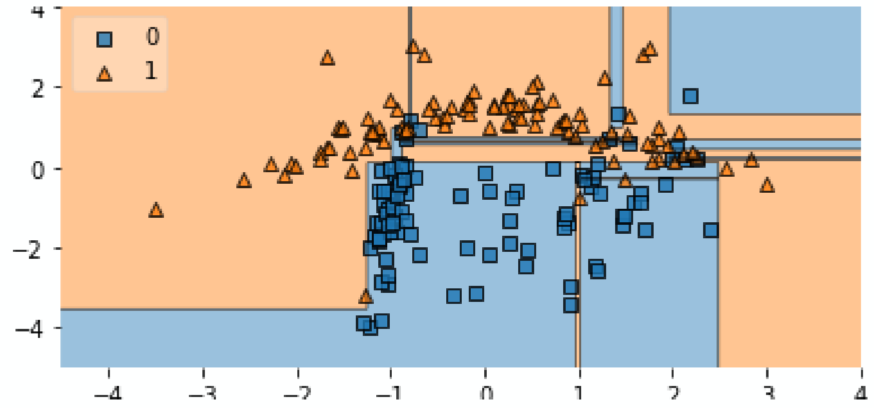
B



Quiz

- Which one depics the decision boundaries drawn by a decision tree?

Decision tree



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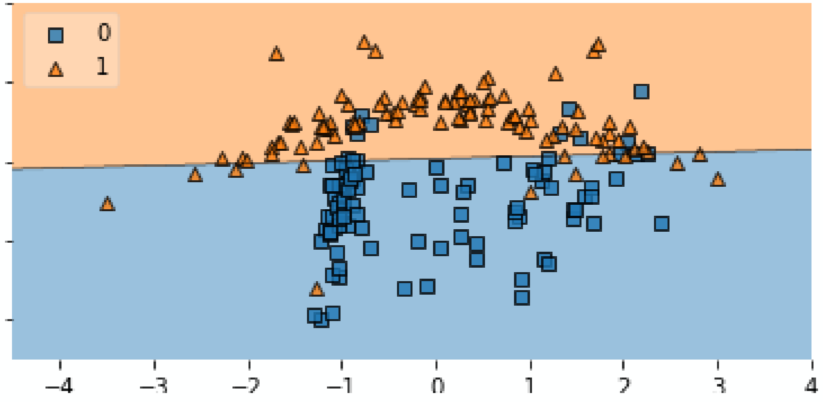
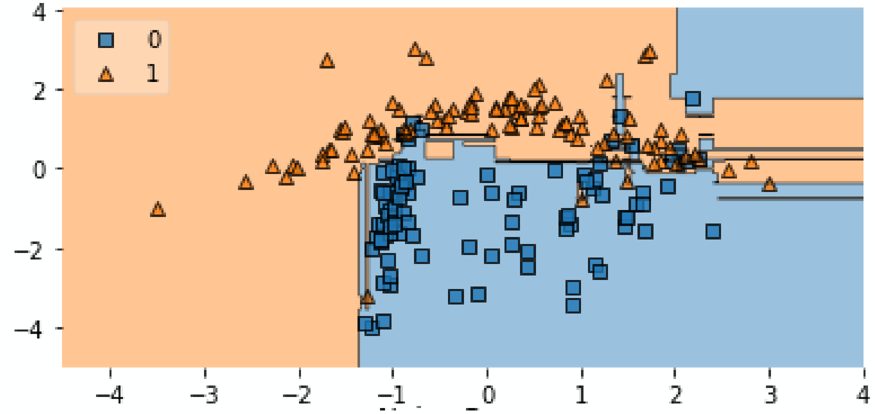


B



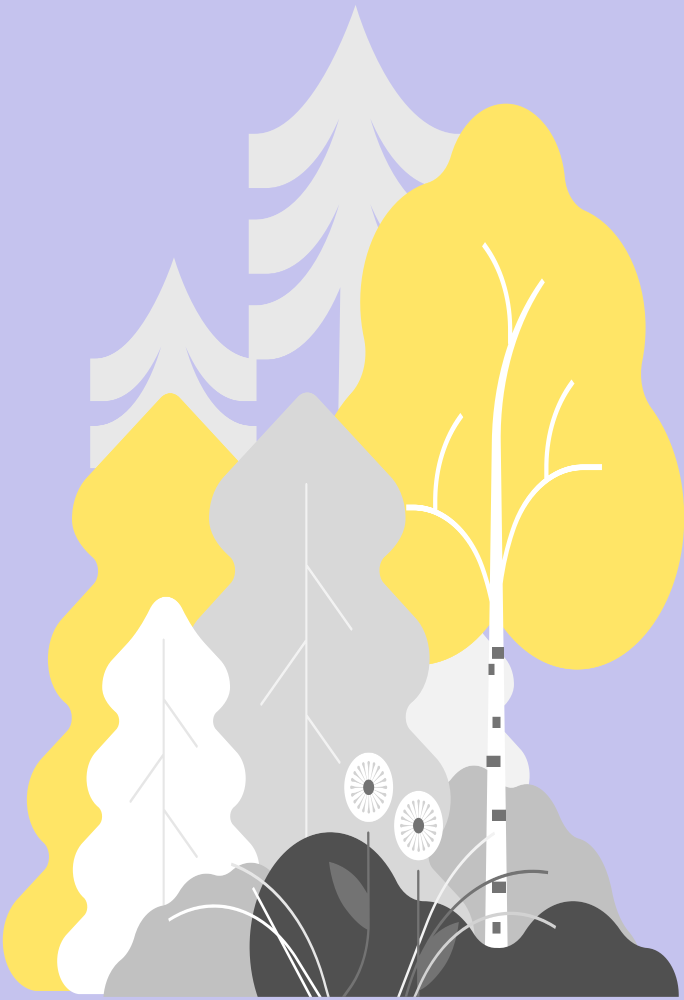
Limitations of Decision Tree

🧮 **Limited to Axis-Aligned Splits:**

Decision boundaries are **rectangular**, may not capture complex patterns well

Supervised Learning

Classification with Random Forest



Random Forests
Ensemble learning with decision trees

Training data

Mass (g)	Color	Texture	pH	Label
84	Green	Smooth	3.5	**Apple**
121	Orange	Rough	3.9	**Orange**
85	Red	Smooth	3.3	**Apple**
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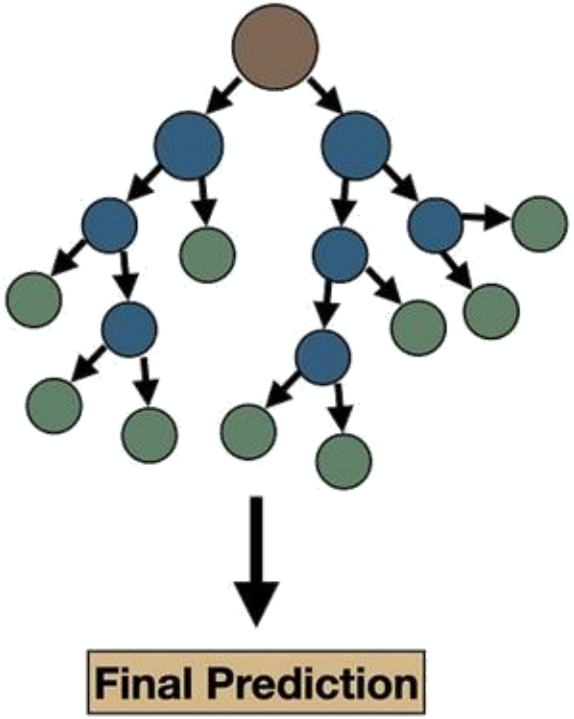
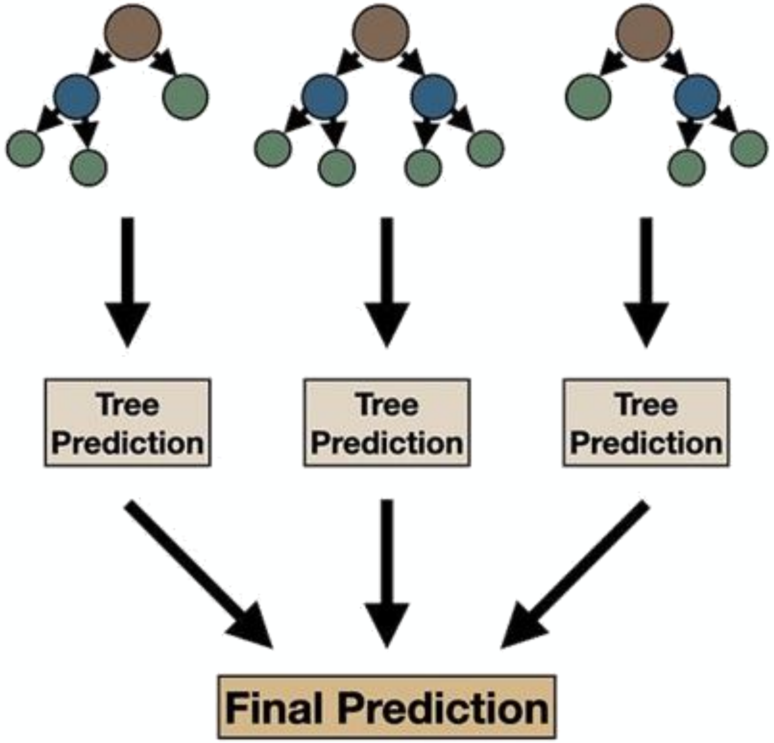
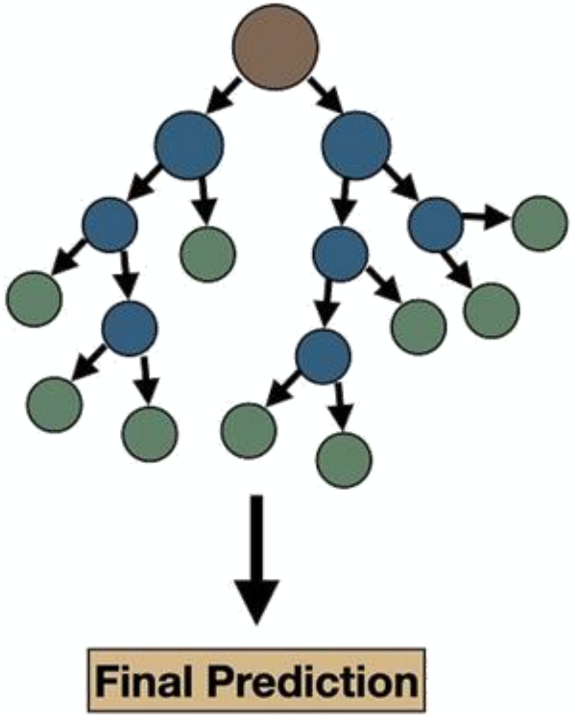
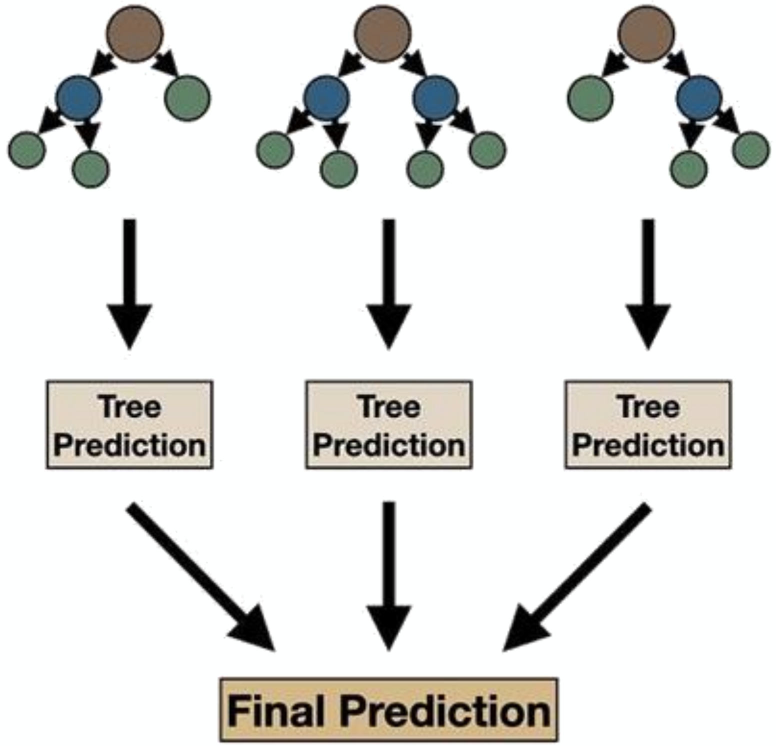
…				
117	Red	Rough	3.4	**Orange**

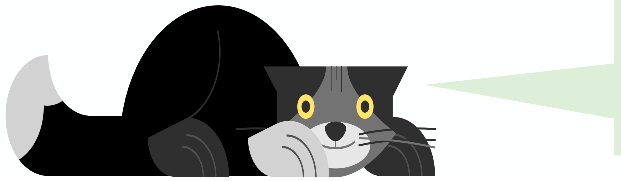




Random Forests
Ensemble learning with decision trees





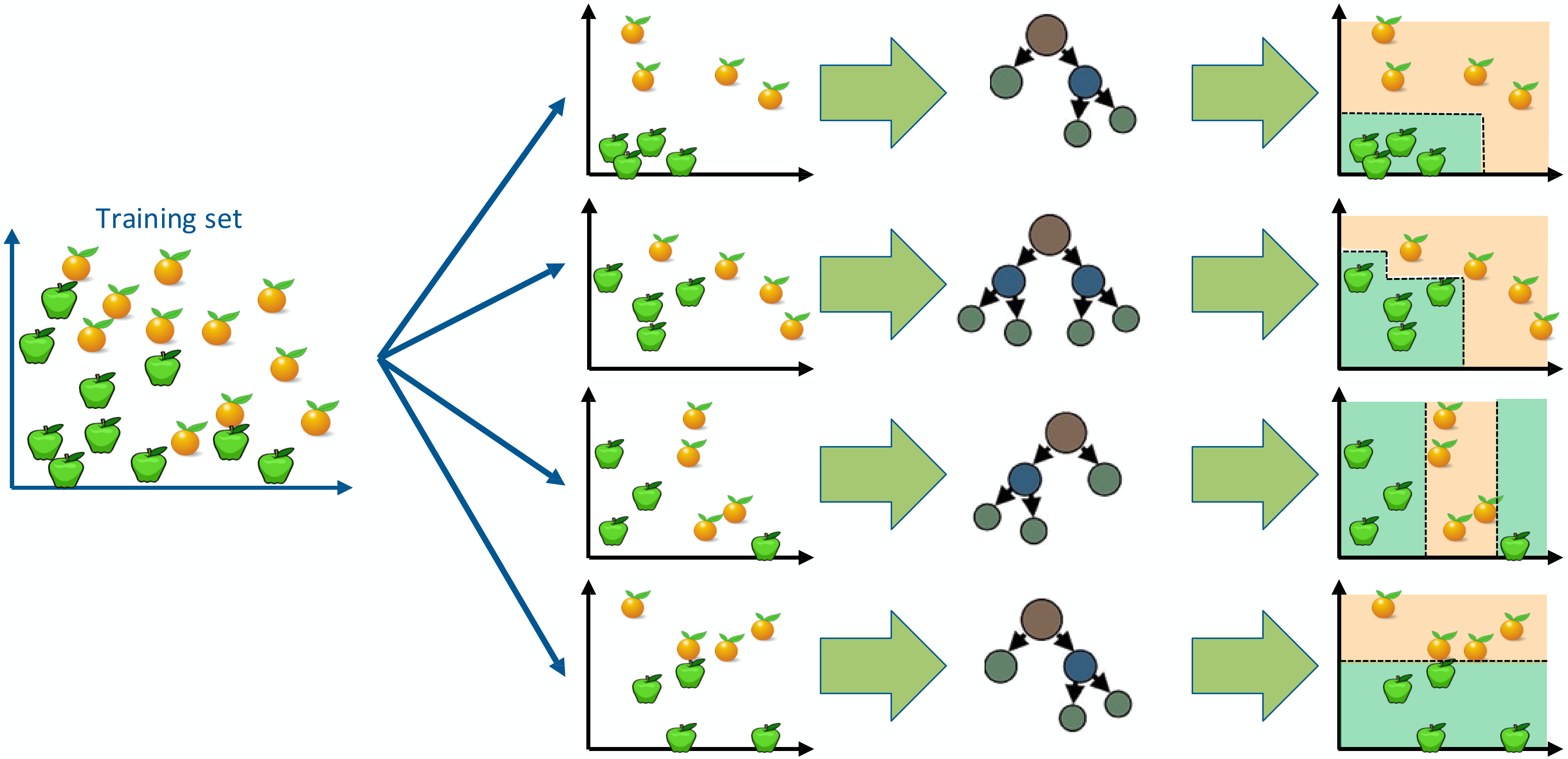


But will these trees learn the same splits (strong pattern) with highest information gain sequence as compared to the single tree?

A: It will. **B: It won't.**

How to ensure diversity: Bagging at training time

M subsets (sampling with replacement)



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How to ensure diversity: Random Subspace Method at training time

Training data

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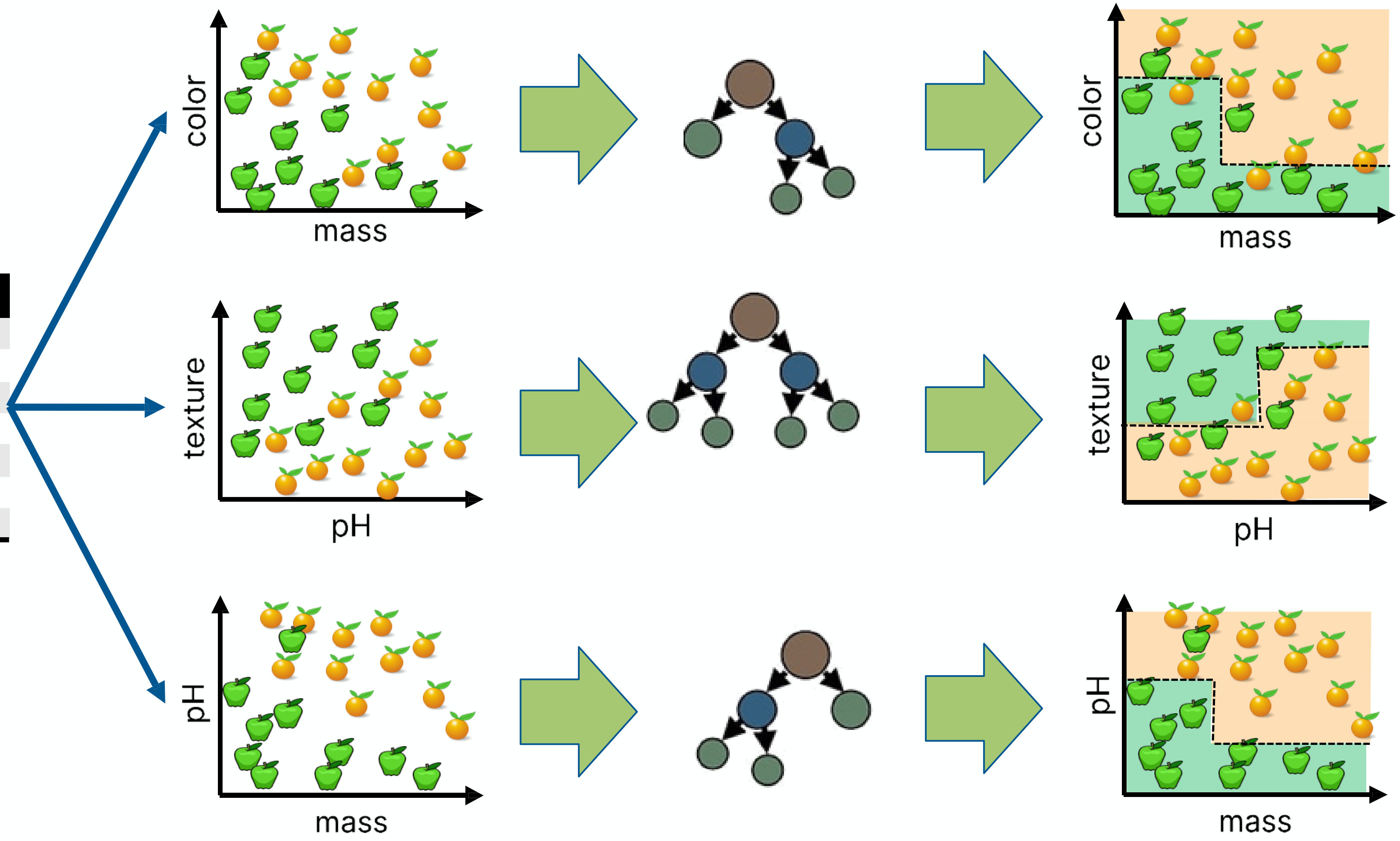
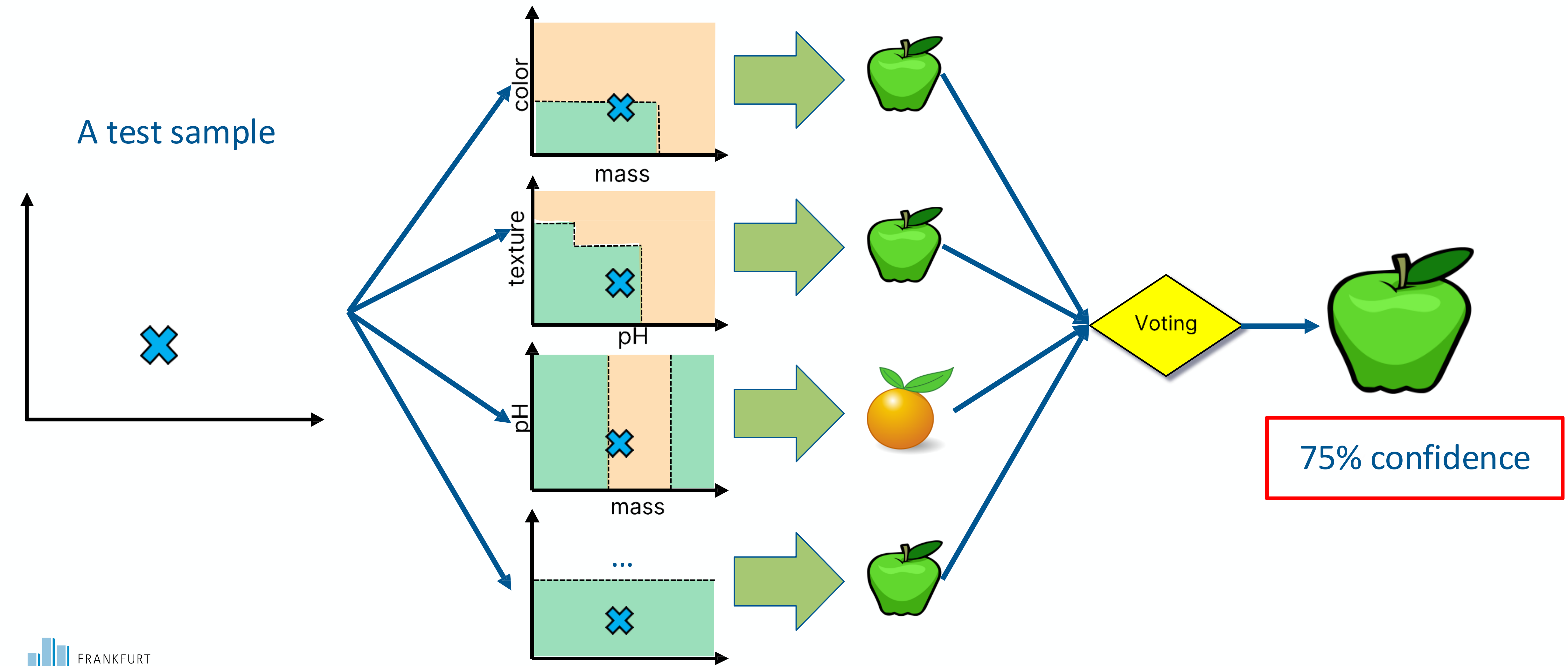


Select a subset of feature

Bagging at inference time



Random Forests

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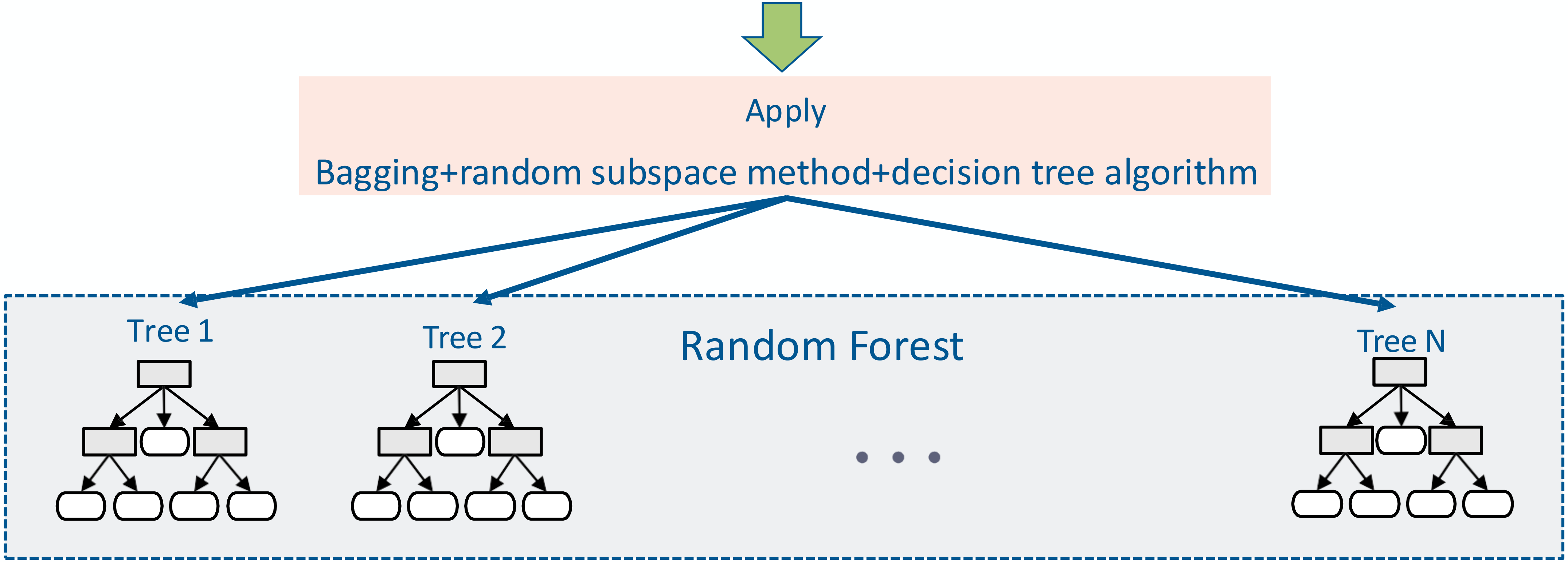
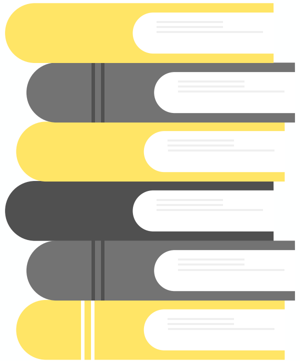


Time for analogy

- Instead of giving 6 books to one student, we assign the task to the whole class. However, to prevent all scholars to learn the same low-hanging fruits, we given each student a subset of the books and within each book a subset of pages.

- **We rely on collective wisdom as it often outperforms a single guess**



Decision Trees and Random Forest (Python)

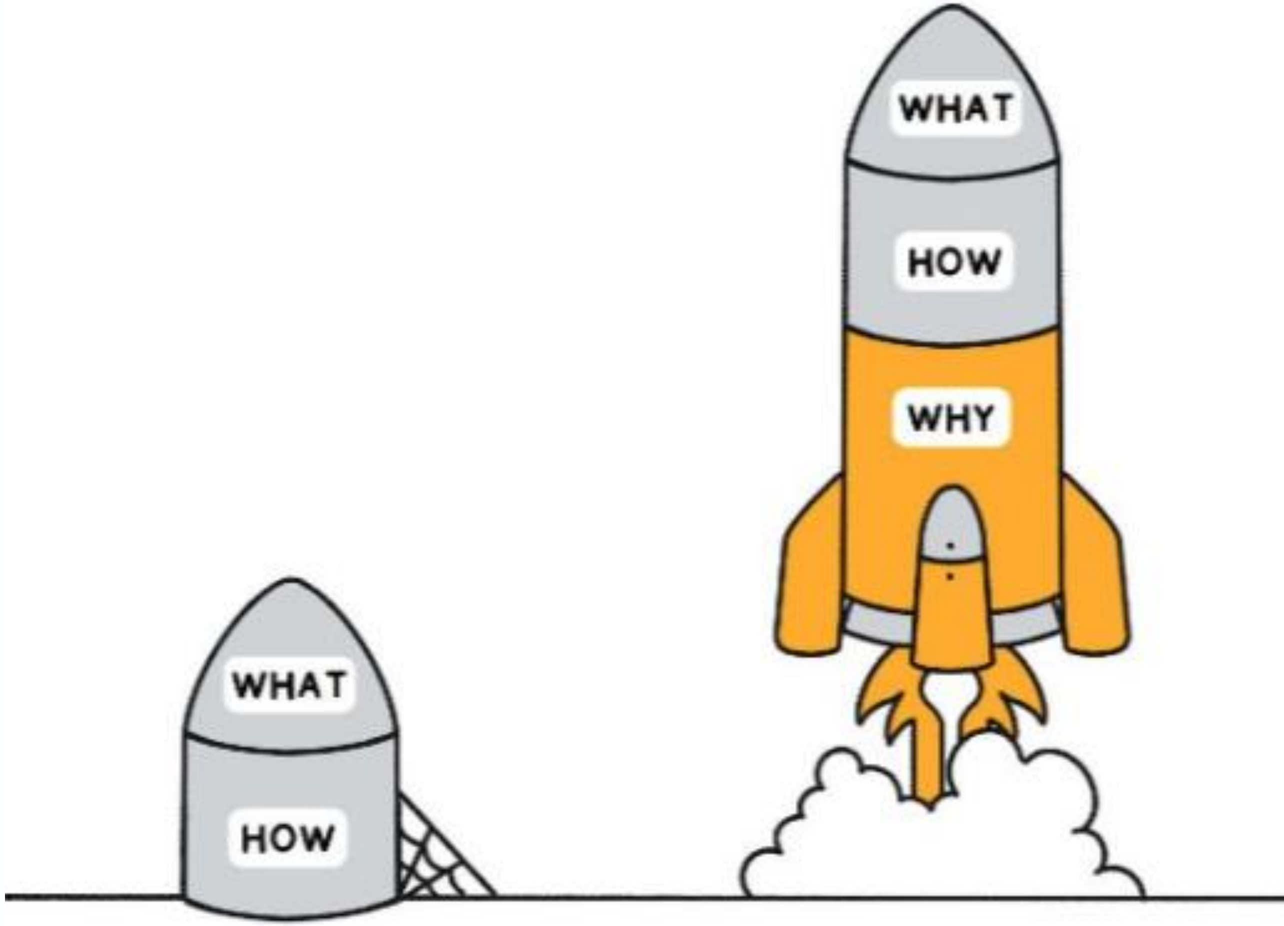
```
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier

clf = DecisionTreeClassifier(criterion = "entropy", min_samples_leaf = 3)
# Lots of parameters: criterion = "gini" / "entropy";
#                     max_depth;
#                     min_impurity_split;
clf.fit(X, y) # It can only handle numerical attributes!
# Categorical attributes need to be encoded, see LabelEncoder and OneHotEncoder
clf.predict([x]) # Predict class for x

clf.feature_importances_ # Importance of each feature
clf.tree_ # The underlying tree object

clf = RandomForestClassifier(n_estimators = 20) # Random Forest with 20 trees
```

Causlity and explainability



Explainable AI (XAI)

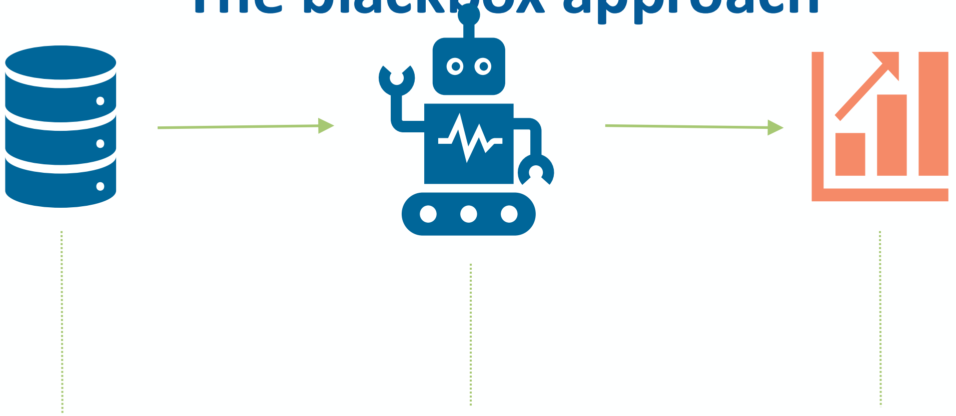
- Explainable AI aims to make **machine learning models more transparent and interpretable**. It helps users understand *how* and *why* a model makes certain predictions, which is essential for trust, accountability, and fairness.
- **Key Concepts**:
  - **Model Interpretability**: Understanding the internal logic of a model's decisions.
  - **Feature Importance**: Identifying which features have the most influence on the model's predictions.
  - **Post-hoc Explanations**: Techniques used after model training to explain the output of complex models like deep learning.
- **Example Use Cases**:
  - **Finance**: Understanding credit scoring models for fairness and compliance.
  - **Healthcare**: Interpreting predictions from diagnostic models for transparency and trust.

Real–world example

- **Scenario**: In a hospital, an AI model diagnoses a patient with vague symptoms and recommends an intensive, high-risk, and costly treatment.
- **Problem without Explainability**:
  - The AI model gives no reasoning behind its recommendation.
  - The doctor and patient are left in the dark—unable to understand the *why* behind the AI's suggestion.
  - Trust is compromised, and the doctor can't fully justify or validate the treatment to the patient.
- **Explainable AI Solution**:
  - The model highlights specific symptoms, risk factors, or patterns it used to arrive at its decision.
  - Doctors can better understand the AI's logic, confirming if the recommendation aligns with medical judgment.
  - Patients gain insight into their treatment, empowering them to make informed decisions and feel confident in their care.
- **Bottom Line**: Explainable AI is crucial in high-stakes fields, as it builds trust, empowers decision-makers, and enables collaborative, transparent, and informed choices.

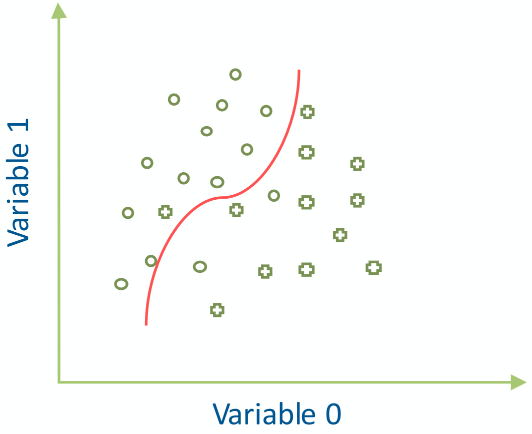
Disease predictions
The blackbox approach

Multi-dimensional
health data
EMR, claims..etc.



Input data **ML Model** **Predictions**

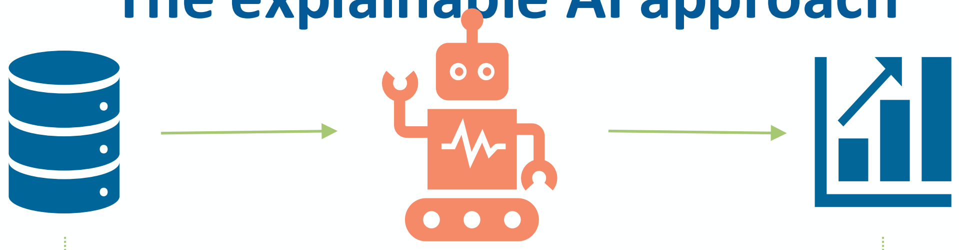
Examples:
- Patients at risk of hospitalization
- Fast progressors in CKD
- Patients at risk of developing Long-COVID



The model learns the risk factors and use them to predict patients with outcome

Disease predictions
The explainable AI approach

Multi-dimensional health data
EMR, claims..etc.



Input data

ML Model

Predictions

Examples:
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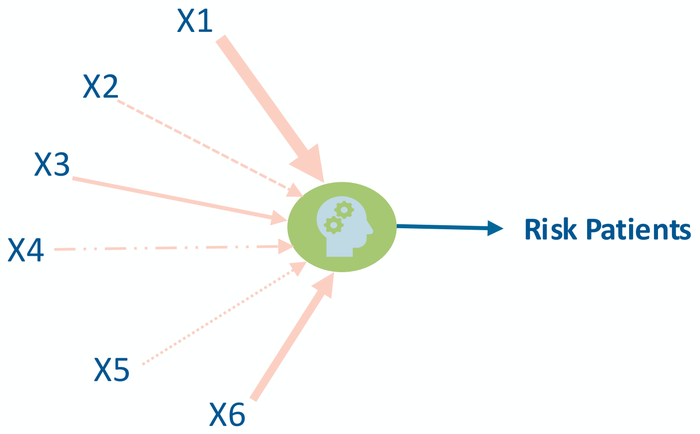

Leverage the model to get the risk factors the model used to predict patients with outcome

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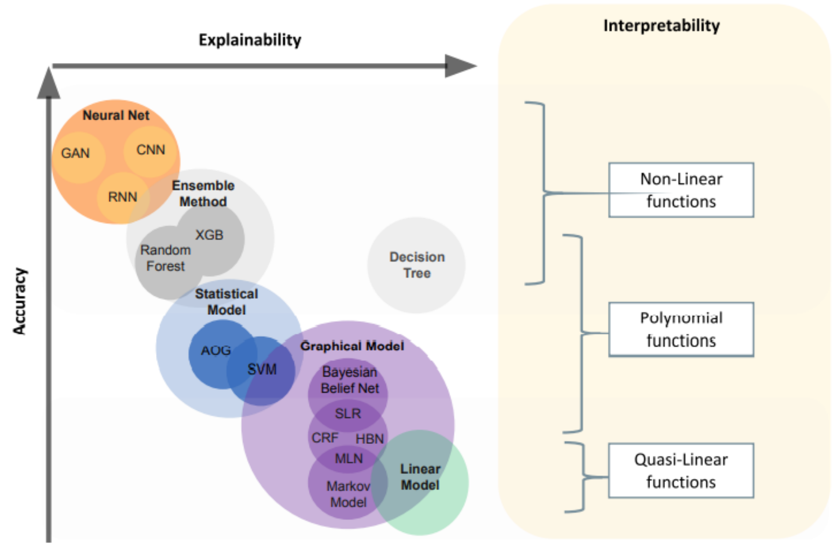
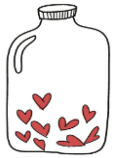
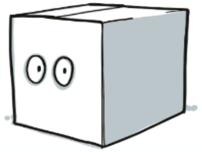
Accuracy vs. Explainability for complex problems!





Glass box models

- Linear regression
- Logistic regression
- Generalized Linear Models
- Decision tree
- K-nearest neighbour
- Naïve Bayes Classifer
- Support Vector Machine in low dimensions



Black box models

- Random forest
- Artificial Neural Nets (MLP, CNN, RNN, etc)
- Gradient boosted models
- Support Vector Machine in high dimensions

Are there ways not to trade off accuracy for explainability? And what are the regulatory thoughts on this?

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Causal AI

- Causal AI focuses on understanding **cause-and-effect relationships** rather than just correlations. It goes beyond predictive models by identifying what *causes* certain outcomes, which is crucial for decision-making, policy setting, and understanding interventions.
- **Key Concepts**:
  - **Causal Inference**: Determining if one event or variable causes another.
  - **Counterfactual Analysis**: Asking "what if" questions to assess outcomes under different scenarios.
  - **Interventions**: Simulating or applying actions to see their causal impact.
- **Example Use Cases**:
  - **Healthcare**: Identifying the effect of a treatment on patient outcomes.
  - **Economics**: Analyzing the impact of policy changes on economic indicators.

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Background and related questions

What is the likelihood this patient with breast cancer, will survive 5 years?

A long survival time may be because of treatment!

What if we must decide to start an aggressive treatment based on the predicted outcome?

Other questions:

- Does smoking cause lung cancer?
- Does obesity cause low response for drug A?
- Pr(lung cancer|smoker) vs Pr(lung cancer|nonsmoker)

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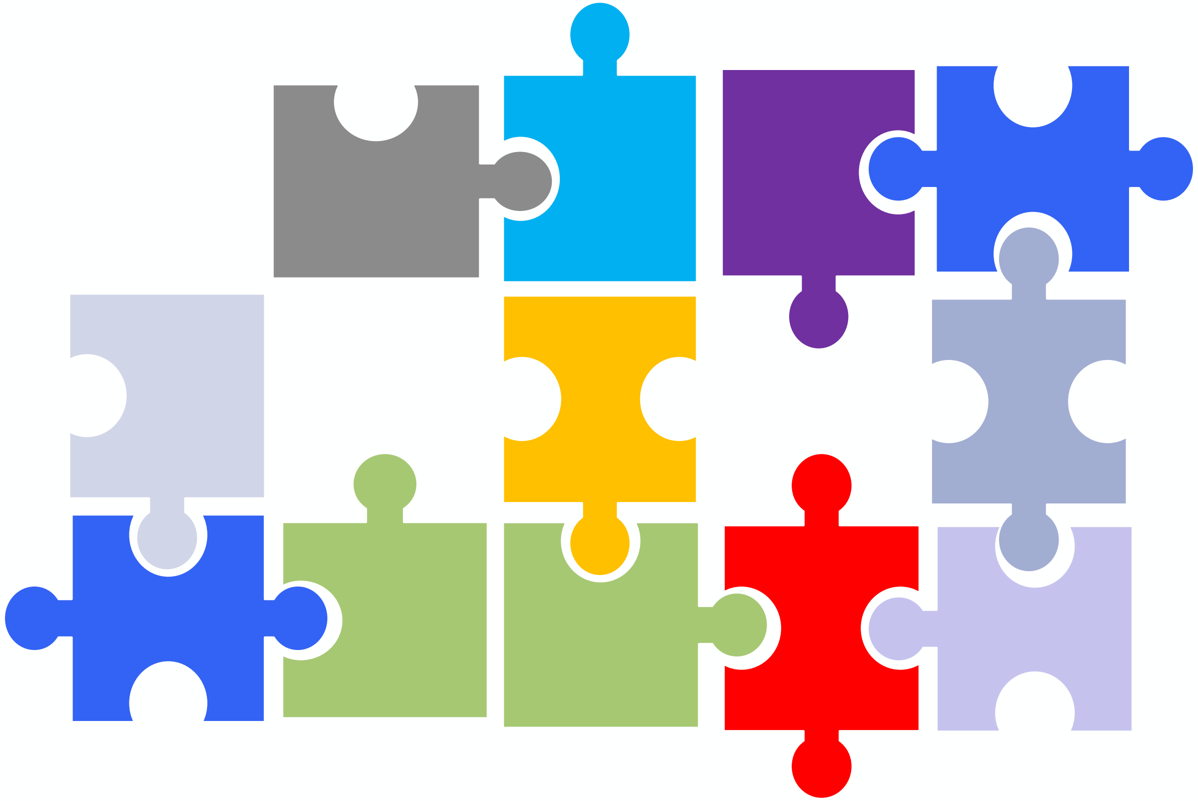
Explainable vs. Causal AI

Explainable AI	Causal AI
• Focuses on making AI models' decisions and processes more understandable to humans. • Provides reasoning behind predictions based on patterns in the data, often using methods like feature importance, attention maps, or decision trees. • Answers the question: *"What factors influenced this particular decision?"*	• Aims to uncover cause-and-effect relationships, going beyond correlation to determine *why* changes in one variable lead to changes in another. • Uses causal inference techniques, often relying on domain knowledge, experiments, or observational data to validate real-world causal links. • Answers the question: *"What would happen if we intervened or changed a specific factor?"*

Questions

Answers



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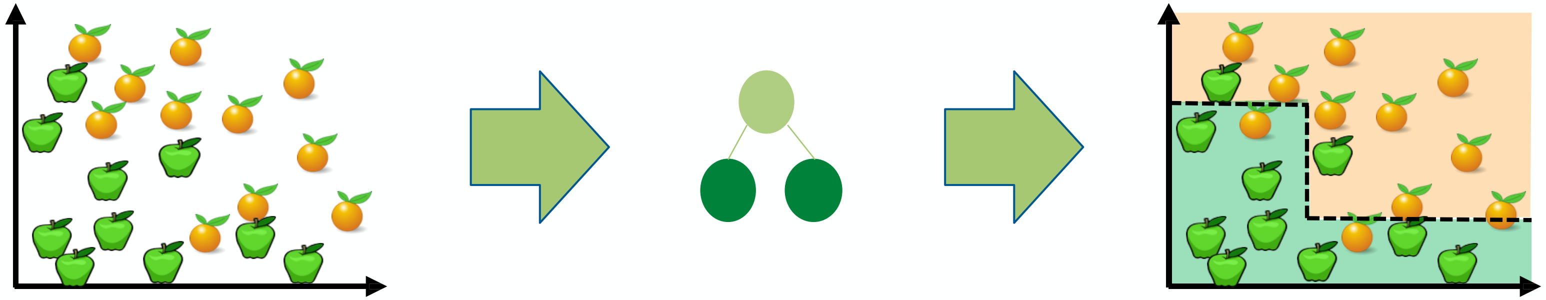
Bonus slides: Supervised Learning

Classification with Adaptive Boosting

Adaptive Boosting : Calculation of weak learner's weight



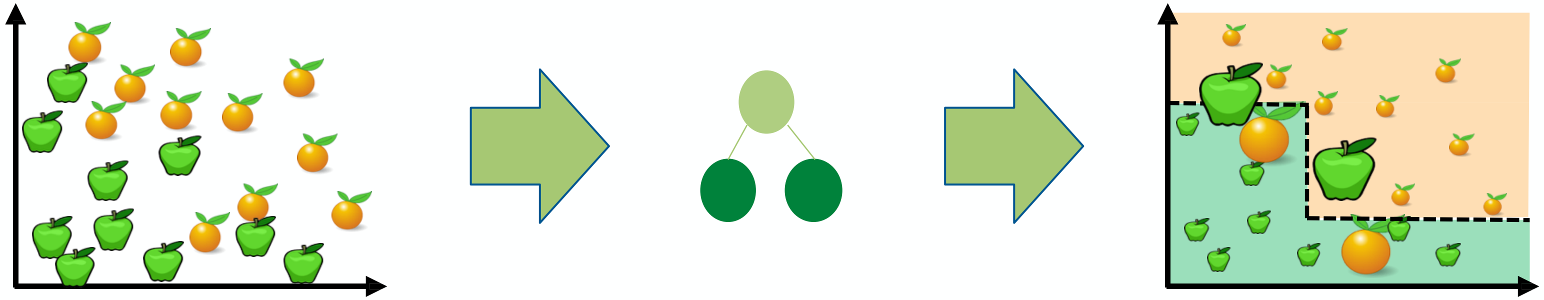
All N samples are assigned equal weight:1/N.
The weights will be used to sample from the training data

Weight Calculation

$$\alpha_i = \frac{1}{2} \ln\left(\frac{1-\varepsilon_i}{\varepsilon_i}\right)$$

ε_i or weighted error rate is calculated as the sum of the weights of the misclassified samples.

Adaptive Boosting: reweighting



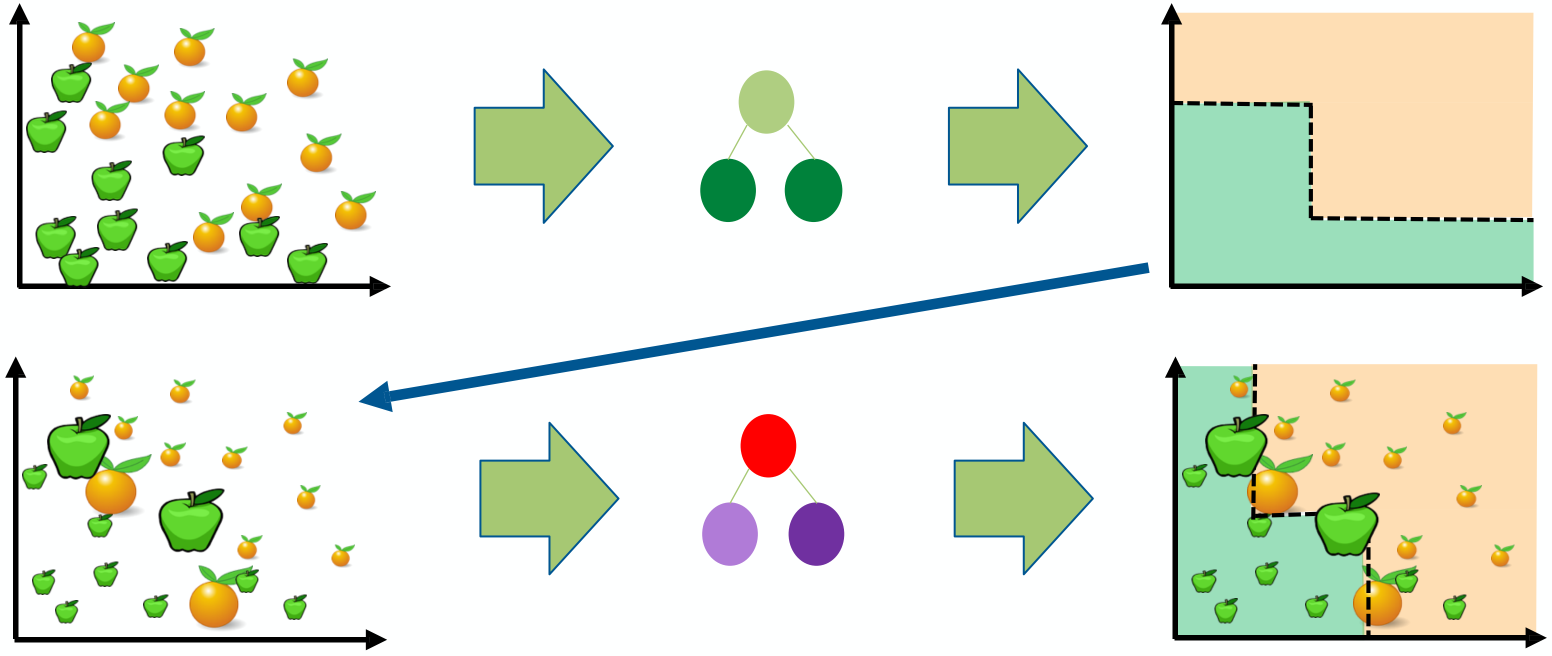
Reweight based on current model's mistakes

$$w_j^{(i+1)} = \frac{1}{Z_i} \begin{cases} w_j^{(i)} \exp^{-\alpha_i} & \text{if } C_i(x_j) = y_j \\ w_j^{(i)} \exp^{\alpha_i} & \text{if } C_i(x_j) \neq y_j \end{cases}$$

where Z_i is the normalization factor

Reweighting is conducted so that decision trees can be grown which favour the splitting of sets of samples with large weights.

Adaptive Boosting: weighted sampling

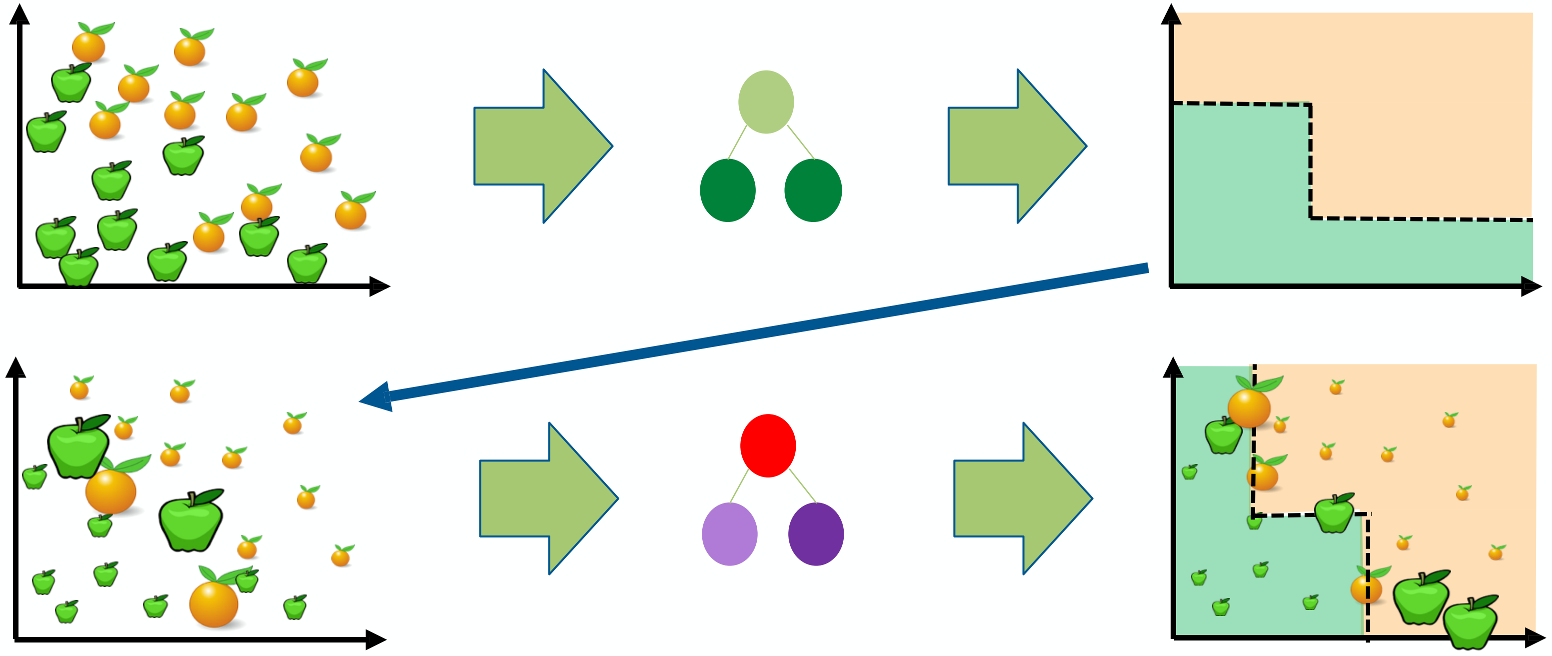


Next model pays more attention to the weighted samples

How?

Weighted Sampling: The dataset is adjusted based on the updated weights. Misclassified samples with higher weights have a greater chance of being selected in this weighted sampling process, making them more likely to appear in the training set for the next weak learner.

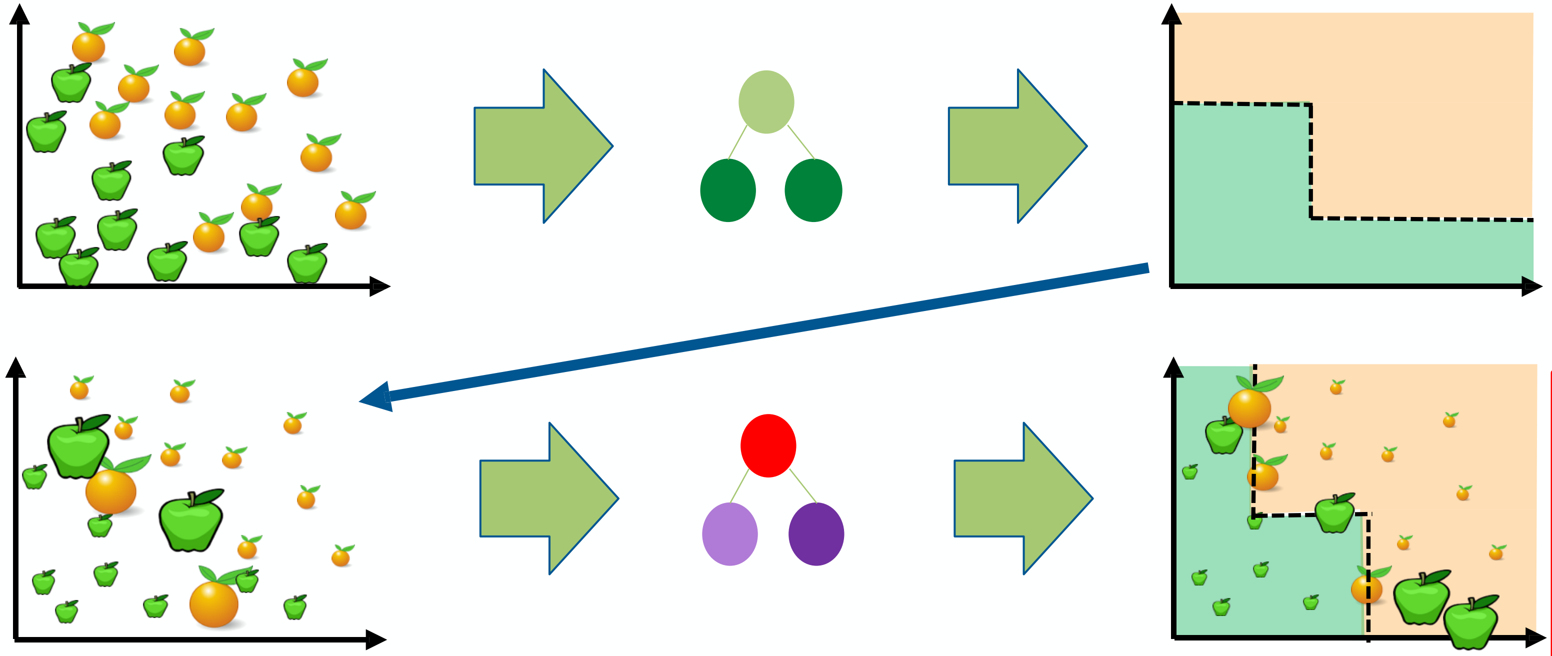
Adaptive Boosting: iteration



$$\alpha_i = \frac{1}{2} \ln\left(\frac{1-\varepsilon_i}{\varepsilon_i}\right)$$

ε_i is calculated as the sum of the weights of the misclassified samples.

Adaptive Boosting: iteration



Reweight based on current model's mistakes

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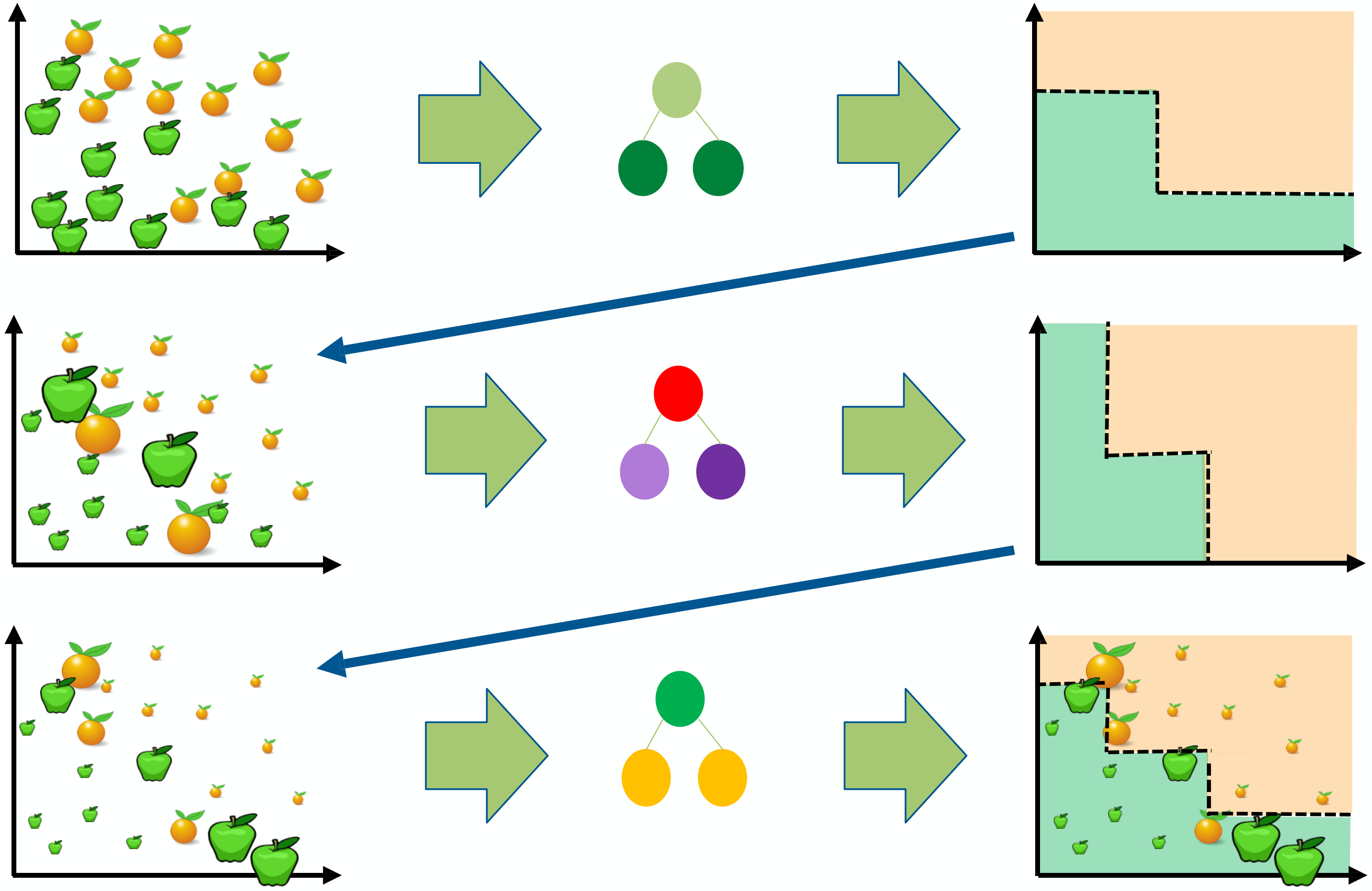
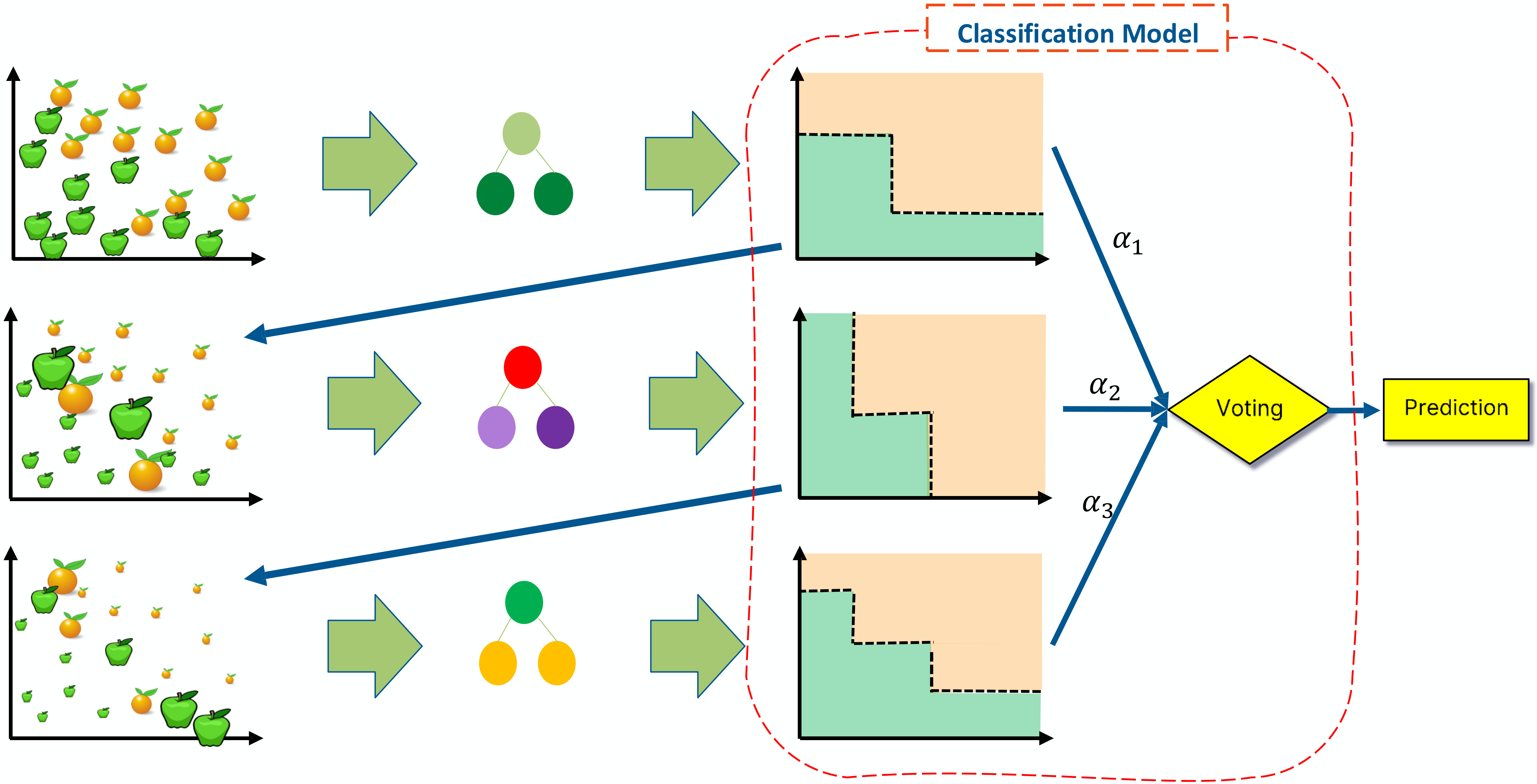
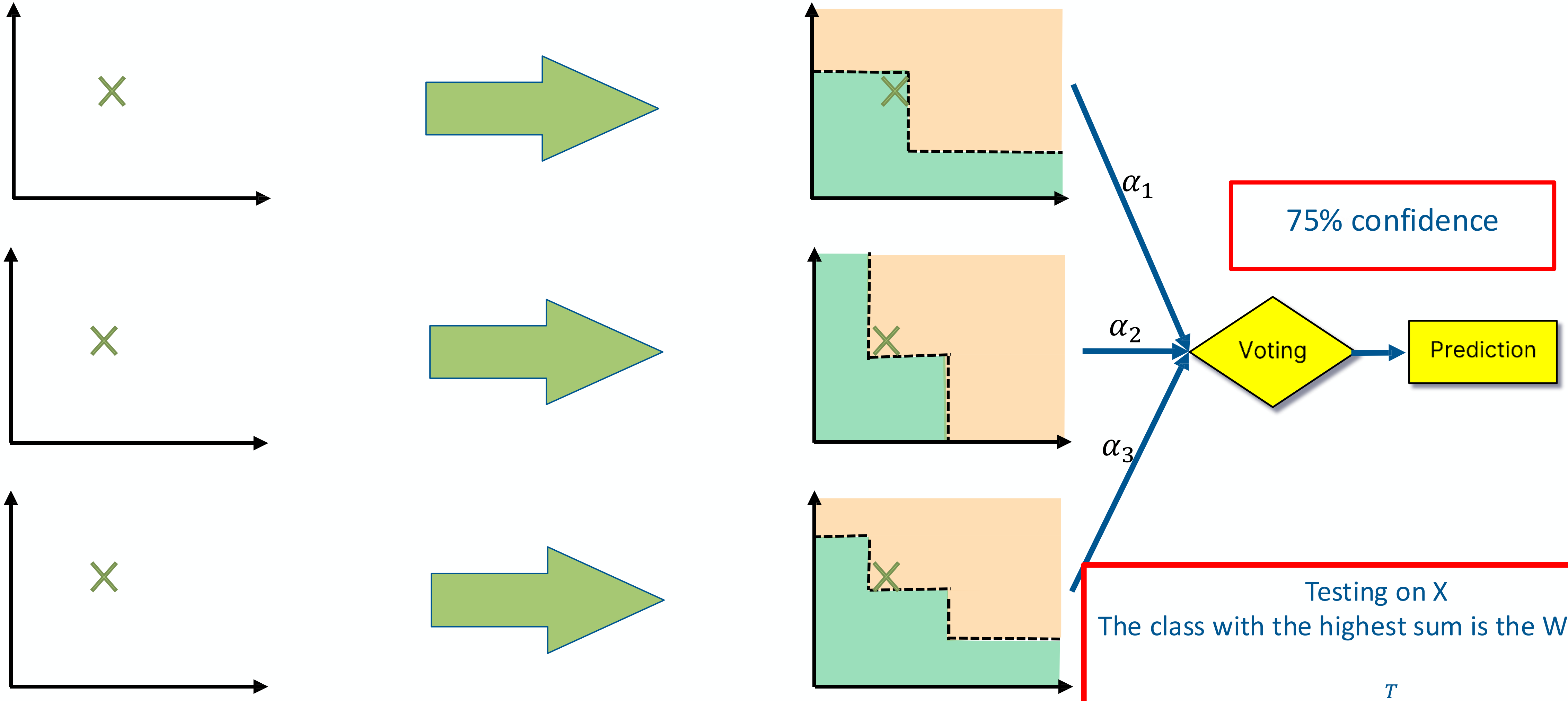
Adaptive Boosting: iteration



Adaptive Boosting



Adaptive Boosting: inference



75% confidence

Testing on X
The class with the highest sum is the WINNER!

$$C*(x_{test}) = \underset{y}{\operatorname{argmax}} \sum_{i=1}^{T} \alpha_i \delta(C_i(x_{test}) = y)$$

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Summary of Ensemble Learning

- **Definition:** Method that combines multiple learning algorithms to obtain performance improvements over its components

Different interesting techniques are applied

- Bagging: multiple models on random subsets of data samples
- Random Subspace Method: multiple models on random subsets of features
- Boosting: train models iteratively, while making the current model focus on the mistakes of the previous ones by increasing the weight of misclassified samples

Adaptive Boosting vs. RDF

	RDF	Adaptive Boosting
Model complexity	Full tees with depth as a hyperparameter	Many “weak learners” - almost always stumps.
computation	Parallel trees: jobs to be parallelized and thus it can handle large datasets more .	Sequential stumps: It takes a step-by-step method.
Sampling	Bagging: each tree gets a subset to learn from independently	Boosting: new learner to improve the shortcomings of existing weak learners.
Combining results	each tree has an equal vote	Some stumps get more say

Adaptive Boosting vs. RDF

Overfitting	Random Forest is generally more robust to overfitting compared to AdaBoost because bagging, which helps to reduce variance and prevent overfitting.
Noise and outliers	AdaBoost is sensitive to outliers since it tries to fit each new model to the misclassified instances of the previous models, which may include outliers. Random Forest, on the other hand, is less sensitive to noise and outliers due to the averaging process.
Imbalanced datasets	Although the AdaBoost algorithm can be directly used to process imbalanced data, the algorithm focuses more on the misclassified samples than samples of minority class. In addition, it may generate many redundant or useless weak classifiers, increasing the processing overhead and causing performance reduction

Homework with Real world example
Comparison between RDF and Adaptive Boosting

- "Imagine we're working for a major streaming service like Netflix or Spotify. Our task is to predict what kind of movies or music a new user will like based on their profile and behavior.
- With Random Forests, we could create a robust model that considers a wide range of factors - from the user's age and location to the time they usually watch movies or listen to music. This could give us a good overall prediction, but it might be hard to explain why the model recommends a specific movie or song.
- On the other hand, with Adaptive Boosting, we could create a model that iteratively learns from its mistakes. It starts with a simple rule - for example, 'people under 25 usually like pop music'. Then it adds more rules to correct the mistakes of the previous ones - 'but if they also follow a lot of indie artists, they might prefer indie pop'. This could give us a highly accurate prediction, but it could also be thrown off by a few unusual users.
- In the end, the best algorithm to choose would depend on our priorities - do we want a model that's robust and handles a wide range of scenarios, or one that's highly accurate but might be sensitive to unusual data?"

Questions

Answers